

Tail Risk at the Last Quote: A Conditional Link Between Two Market Papers

Research phase: PAUSE after 1 round, 0 ITERATE, 0 REVISE. Edit phase: readable note

Abstract

Struski and colleagues study language-model traders in a double auction, while the companion local manuscript studies fixed-batch estimates of conditional value at risk in a different allocation mechanism. We asked whether the sampling issue in the second paper could change a late trader's choice between taking a sure profit and waiting for a better one. The peers found a valid reversal for a constructed auction state: an expected one-sample risk estimate favours waiting, while true tail risk favours trading now. The continuation behaviour and the fill probability were assumed, and no such state was shown to occur in the reported runs. The thread therefore paused pending a matched-state check in the auction simulator.

1 Introduction

Competitive Market Behavior of LLMs puts language-model buyers and sellers in a double auction [3]. A buyer may accept the best current selling price, or post a better buying offer and hope someone accepts it later. Each round has at most 300 chances to act, so a better offer can expire without a trade. The agents are told to maximise profit. The paper reports that one model often makes tiny price improvements and leaves profitable trades unfinished; its markets reach lower efficiency than the best-performing model's markets.

Fixed-Batch CVaR and Participation in a Quadratic Resource-Allocation Mechanism studies a separate allocation game [2]. It measures risk with conditional value at risk, or CVaR: the average loss in the worst chosen fraction of outcomes. Its main example shows that optimising a fixed-batch estimate, even in expectation over fresh batches, can lead to a choice that fares worse under true risk. Its equilibrium and welfare bounds require the assumptions of its quadratic mechanism. The two papers called "Q" and "P" inside that manuscript are other works, not the pair considered here.

The connection pursued was a late auction choice. Accepting a standing offer secures a modest gain; improving one's own offer may yield a larger gain, but may never be filled. The peers asked whether fixed-batch CVaR could rank those actions differently from true CVaR. This was a counterfactual question: the auction paper gives its traders no CVaR objective or sampling oracle.

2 What was done

The peers first read the auction rules and the risk paper's distinction between true CVaR and the *expected* CVaR of an empirical batch. Ada also asked whether the auction paper's means across ten runs could reveal the downside of market efficiency. They cannot: different sets of run outcomes can have the same mean and very different worst outcomes. This raised a separate question about market-wide risk.

Emmy then reduced the trading question to two actions at the end of a round. Crossing the standing offer gives a sure profit. Posting a noncrossing offer gives a larger profit only if a later trader accepts it. Emmy derived the risk ranking for this choice and placed it in a state allowed by the auction rules. Ada checked the seller count, the resulting fill probability, and the surplus arithmetic against those rules. Their derivation and checks are in `../emmy/cross_wait.md` and `../ada/cvar_cda_note.tex`.

The peers next tested the interpretation. They separated an expected batch objective from a decision based on one realised batch, and checked what information the selected trader can actually see. They also checked whether a local decision result says anything about total market surplus. These checks narrowed the result to a conditional example.

3 Results

Let crossing yield certain profit $g > 0$. Let a specified continuation after posting a better, noncrossing offer yield profit $h > g$ with probability p , and zero otherwise. Write $f = 1 - p$ for the chance of no trade, and let β be the fraction of worst losses used in CVaR. A trader's risk-adjusted utility is the negative CVaR of its loss, where loss is negative profit. Directly averaging the worst β fraction gives the following utility for waiting:

$$U_{\text{wait}} = h \max \left\{ 0, 1 - \frac{f}{\beta} \right\}.$$

Crossing has utility g . With one fresh observation, the *expected empirical* CVaR gives waiting utility ph , just its expected profit. Thus, if $f \geq \beta$ and $ph > g$, the expected one-sample objective selects waiting while true CVaR selects crossing. This is an action-ranking result under the stated continuation, not a claim about how the observed language models choose.

The constructed state makes the comparison concrete. At the penultimate iteration, suppose all 22 traders remain untraded. A buyer values a unit at 3.25, the best ask is 3.00, and the buyer can instead improve the best bid to 2.25. Crossing gives $g = 1/4$; if a seller accepts the new bid next, the buyer earns $h = 1$. Seven sellers in the auction's value grid have costs at most 2.25. Stipulate that each of them crosses if selected next and that every other selection leaves this buyer without a trade. Uniform selection then gives $p = 7/22$ and $f = 15/22$. For a worst-half tail, $\beta = 1/2$, the checked rankings are

$$U_{\text{wait}} = 0 < \frac{1}{4}, \quad U_{\text{one sample}} = \frac{7}{22} > \frac{1}{4}, \quad U_{\text{two samples}} = \frac{49}{484} < \frac{1}{4}.$$

The two-sample value follows because, for this tail size, both observations must be successful for waiting to have positive empirical risk utility. These are expectations over fresh batches. A single realised one-sample estimate instead favours waiting after a sampled fill and crossing after a sampled failure. Emmy derived the batch calculation; Ada independently checked the auction count and inequalities in the peer files named above.

If the standing ask belongs to a seller with cost 1, crossing creates surplus $9/4$. Under the same stipulated continuation, waiting creates expected surplus $49/88$, as Ada checked. This comparison concerns only that constructed state. Its waiting option has true-risk utility zero, the same as no trade, so the reversal is not the negative-participation result in the risk paper.

4 Objections and unresolved points

The fill probability assumes a policy for every next trader and uses private seller costs. The auction buyer sees only its own value, the book and histories. The peers therefore used 7/22 as an analyst's counterfactual, not a trader's belief. A practical test needs an oracle or a belief from public history. No supplied evidence shows that this state occurred, or that the auction agents optimised CVaR; here the one-sample choice also maximises expected profit.

Two market objections remain. A buyer who crosses can displace a higher-value buyer, so more crossing need not raise surplus. Separately, the sum of individual CVaRs need not equal CVaR of aggregate loss. The auction reports ten-run means, which do not reveal downside efficiency: ten values of 0.91 and nine values of 1 plus one of 0.1 share a mean but have different worst runs. These invented sets do not reproduce the auction's data.

The quadratic taxes, equilibrium assumptions and value-error bounds of the risk paper do not carry over to this discrete auction decision. The auction's ten evaluation runs are also different from the number of observations in a trader's CVaR batch. The ledger's outside sources already cover risk-based continuous double-auction bidding [5], CVaR bidding in a double-auctioned power market [1], and sampling bias in CVaR [4]. The recorded searches do not establish priority for the proposed combination.

5 Why the thread stopped

The verifier accepted the ranking reversal but paused the thread because it rests on assumed continuation behaviour and a fill probability unavailable to the traders. No evidence shows that relevant states occur or change market outcomes. The smallest restart is to count reachable late states in the auction simulator and estimate action-specific continuation distributions at matched states with fixed other-trader policies. Only then could tests of public-history beliefs and market outcomes support a claim about these agents.

References

- [1] Panda et al. CVaR bidding in a double-auctioned power market, 2019. URL: <https://doi.org/10.1049/iet-gtd.2018.5905>.
- [2] Pathfinder research campaign. Fixed-Batch CVaR and Participation in a Quadratic Resource-Allocation Mechanism, September 2026. Local manuscript, pathfinder-local-S4.
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- [4] Tamar et al. Sampling bias in CVaR optimisation, 2015. URL: <https://doi.org/10.1609/aaai.v29i1.9561>.
- [5] Vytelingum et al. Risk-based continuous double-auction bidding, 2004. URL: <https://eprints.soton.ac.uk/259567/>.